Anjan Nepal

State College, PA • apn5663@psu.edu • (502)319-0443
https://orcid.org/0009-0008-1554-0647

EDUCATION

PhD, Agriculture and Environmental Plant Science
Pennsylvania State University, State College, PA
2025-Present
Focus: Anaerobic soil disinfestation, organic vegetable production, weed management

MS, Environmental Studies. GPA: 3.87
Kentucky State University, Frankfort, KY
2021-2023
Thesis: Testing Bioaccumulation of Heavy Metals in Cabbage and Mustard Grown under Five Soil Amendments
Student Team Project: Development of Educational Materials for Addressing Lead in Drinking Water in Schools and Childcare Facilities

B.Sc., Agriculture. GPA: 3.95
Tribhuvan University, Bhairahawa, Nepal
2014-2018
Thesis: Evaluation of Coriander (*Coriandrum sativum* L.) Varieties for Growth and Yield Parameters

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (PhD)
Pennsylvania State University, State College, PA
June 2025- Present
Advisors: Dr. Francesco Di Gioia, Dr. Caio Brunharo

- Design and execute research projects on anaerobic soil disinfestation and weed suppression in the high tunnel vegetable production system
- Collaborated with research mentors, farmers, graduate students, and academics from other institutions to refine research methodologies

Research Associate
Kentucky State University, Frankfort, KY
July 2023 – June 2025
Advisor: Dr. George F. Antonious

- Designed and implemented research on soil amendments, yield and quality of vegetable crops, heavy metal bioaccumulation, and quantification of glucosinolates in *Brassica* species
- Conducted fieldworks: incorporated soil amendments, managed crop cultivation, collected and analyzed water and soil samples using ICP-OES
- Collaborated with research mentors, farmers, graduate students; authored and co-authored peer-reviewed publications

Graduate Research Assistant (MS)
Kentucky State University, Frankfort, KY
August 2021 – May 2023

Advisor: Dr. George F. Antonious

- Conducted fieldwork and laboratory analysis: soil amendments, water sampling, pH/nutrient/heavy metal analysis, ICP-OES operation

PUBLICATIONS

- Neupane, B., Antonious, G. F., Dawood, M. H., & **Nepal, A.** (2025). Bioaccumulation of heavy metals in hot pepper grown in biochar and animal manure-amended soil. *Environmental Pollution and Management*. https://doi.org/10.1016/j.epm.2025.09.006.
- Antonious, G. F., **Nepal, A.**, & Neupane, B. (2025). Remediation of Cd, Cu, and Zn Metals in Soil Amended with Biochar and Animal Manure Using a Hyperaccumulator. *Pollutants*, *5*(3), 22. https://doi.org/10.3390/pollutants5030022
- Neupane, B., Antonious, G. F., Nzaramyimana, T., Acharya, S., Oli, D., **Nepal, A.**, Dhakal, A., & Makumbe, H. H. (2025). Advancement in agriculture through radioisotopes: Current context, challenges and future direction. *Journal of Agriculture and Food Research*, *13*, Article 101966. https://doi.org/10.1016/j.jafr.2025.101966
- **Nepal, A.**, Antonious, G. F., Bebe, F. N., Webster, T. C., Gyawali, B. R., & Neupane, B. (2024). Heavy Metal Accumulation in Three Varieties of Mustard Grown under Five Soil Management Practices. *Environments*, *11*(4), 77. https://doi.org/10.3390/environments11040077
- **Nepal, A.**, Antonious, G. F., Gyawali, B. R., Webster, T. C., & Bebe, F. (2024). Assessing the bioaccumulation of heavy metals in cabbage grown under five soil amendments. *Pollutants*, *4*(1), 58-71. https://doi.org/10.3390/pollutants4010005
- Antonious, G.F., Chiluwal, A.C. and **Nepal, A.** (2023). Chitin, Biochar, and Animal Manures Impact on Eggplant and Green Pepper Yield and Quality. *Agricultural Sciences, 14*, 368-383. https://doi.org/10.4236/as.2023.143024
- Duwal, A., **Nepal, A.**, Luitel, S., Acharya, S., Pathak, R., Poudel, P. R., & Shrestha, J. (2019). Evaluation of coriander (Coriandrum sativum L.) varieties for growth and yield parameters. *Nepalese Journal of Agricultural Sciences*, *18*, 36-46.

PEER REVIEW ACTIVITY

- Pre-publication reviewer, *Advances in Weed Science* (Brazilian Weed Science Society), 2025. Reviewed a manuscript on morphological and molecular identification of weed species
- Pre-publication reviewer, *Air, Soil and Water Research* (SAGE Publications; indexed in Clarivate ESCI, Scopus, DOAJ), 2025. Reviewed manuscript on trace metal contamination in soil and native maize through two revision rounds

SCIENTIFIC CONFERENCES

- Nepal, A*., Negron, R., Nepal, J., Ono-Raphel, J., Brunharo, C., Di Gioia., F. (2026). Effects of Carbon and Nitrogen Amendments on Yield and Mineral Profile of Organic Tuscan Kale Grown in High Tunnel Following Anaerobic Soil Disinfestation. ASHS Annual Meeting, Dallas, Texas. [**Oral Presentation**]
- Nepal, A*., Negron, R., Nepal, J., Ono-Raphel, J., Brunharo, C., Di Gioia., F. (2026). Impact of Anaerobic Soil Disinfestation on Weed Suppression in an Organic Vegetable High Tunnel System. ASHS Annual Meeting, Dallas, Texas. [**Oral Presentation**]

- Nepal, A*., Antonious, G. F., & Dawood, M. H. (2024) Yield, Quality, Disease Symptoms and Heavy Metal Accumulation in Potato Grown Under Five Soil Amendments. [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX. [**Oral Presentation**]
- Nepal, A*., & Antonious, G. (2024, April 6-9). Bioaccumulation of Heavy Metals in Mustard Grown under Five Soil Management Practices. [**Oral presentation**]. ARD 2024: Gaylord Opryland Hotel, Nashville, TN, United States.
- Nepal, A*., Antonious, G. F., Gyawali, B., Webster, T., & Bebe, F. (2023) Bioaccumulation of Copper, Molybdenum, and Zinc in Cabbage Grown Under Five Soil Amendments [**Oral Presentation**]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO.
- Nepal, A*. & Antonious, G.F. (2022). Analysis of Animal Manures Used for Growing Field-Grown Vegetables [**Poster Presentation**]. 20th Biennial 1890 Association of Research Director Symposium. Atlanta, GA.
- Nepal, A*., & Antonious, G. F. (2022) Yield and Quality of Field Crops Grown in Various Soil Amendments [**Poster Presentation**]. ASA, CSSA, SSSA International Annual Meeting, Baltimore, MD.
- Nepal, A*. & Antonious, G.F. (2022). Yield and Quality of Tomato, *Solanum lycopersicum*, Grown in Various Soil Amendments [**Oral Presentation**]. 2022 Annual Meeting, Kentucky Academy of Sciences, Morehead State University, KY.

SKILLS

- **Technical skills:** ArcGIS pro, R statistical software, ICP-OES operation, Microsoft Office Suite
- **Analytical skills:** Statistical analysis, data interpretation, literature review
- **Research skills:** Quantitative research, experimental design, heavy metal digestion
- **Communication skills:** Presentations, academic writing, team collaboration

TRAINING AND CERTIFICATES

- Broadening Extension through Student Training (BEST) May 2026
- 2024 International Soil Health Certificate Course: Cornell University
- Introduction to Probability and Data with R: Coursera 2024
- Python for Data Science, AI & Development (By IBM): Coursera 2025
- How to Write and Publish a Scientific Paper (Project-Centered Course)

PROFESSIONAL AFFILIATIONS

- Student member, Crop Science Society of America (2022-Present)
- Student member, Soil Science Society of America (2022-Present)
- Student member, Agronomy Society of America (2022-Present)
- Kentucky Academy of Sciences (2022-Present)
- American Society of Horticultural Sciences (2026- Present)

AWARDS AND SCHOLARSHIPS

- Walter Thomas Memorial Scholarship 2026/27, College of Agricultural Sciences, Penn State University
- FOCCE (Forest Owner Carbon and Climate Education) Scholars Program 2026

- Golden Jubilee Scholarship provided by the Indian Embassy for bachelor's level of study at Tribhuvan University (2014-2018)